

Artificial Intelligence and Cloud Architecture Report

1. Overview of Project Nova

Project Nova is an enterprise-grade cloud native system designed to automate large-scale data processing workflows. Launched in Q1 2025, the system integrates advanced machine learning models with serverless infrastructure to achieve high scalability and fault tolerance.

The primary objective of Project Nova is to minimize operational latency in cross-border real-time data ingestion pipelines, serving financial institutions across three continents.

2. Technical Infrastructure and Architecture

The infrastructure relies heavily on Kubernetes clusters deployed across multiple regions to ensure high availability. The core computational layer utilizes distinct microservices built with FastAPI and Go, which communicate asynchronously via distributed event streams.

Data persistence is managed using a combination of a highly scalable distributed NoSQL database for time-series data and a vector database for semantic indexing.

3. Security Matrix and Access Control

Security in Project Nova is strictly enforced through an advanced Zero Trust Network Architecture (ZTNA). Every request traversing the API gateway must undergo cryptographic verification and dynamic context-aware authorization evaluations.

Data is encrypted at rest using AES-256 standards, and transport layer encryption relies exclusively on TLS 1.3 to mitigate modern network-based interception vectors.